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Blockchain-enabled secure land record management system with a new lightweight cryptosystem based on a hybrid chaotic map

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ABSTRACT

Land is a fundamental resource with significant economic and social value. Land transactions are often carried out by individuals to fulfill their specific needs and requirements that are usually maintained by different governmental offices. However, a land record (LR) generally contains sensitive information associated with the land. In recent decades, the shift from traditional paper-based approaches to digital systems has become increasingly prevalent. Moreover, this transition has brought about certain challenges, including the potential for single-point failures and security vulnerabilities. A blockchain-powered land record management system (LRMS) has the potential to effectively address and resolve the current challenges while maintaining robust security measures. To design a blockchain-based secure LRMS, this paper develops a new lightweight cryptosystem based on a newly proposed hybrid chaotic map that ensures LR privacy. In addition, the paper also proposes a dynamic way of character to integer mapping scheme named the dynamic character to integer (C2I) table that enhances encryption complexity while reducing conversion overhead by approximately 33% compared to the ASCII table. Besides, it streamlines the process of ownership transfer by facilitating a decentralized land trading platform. Moreover, LR integrity is maintained by incorporating both InterPlanetary File System (IPFS) and blockchain as storage. Finally, the experimental results, analyses, and comparisons indicate the effectiveness of the proposed LRMS over the state-of-the-art systems.

Keywords:

Land record, Land record management system, Blockchain, Lightweight cryptosystem, IPFS.

1. Introduction

Land is an integral and valuable resource worldwide, forming the foundation for human existence and development. Each parcel of land is uniquely identified through various methods such as dag number (khasra number), plot number, parcel number, cadastral number, lot number, block number, section number, geographic information system coordinate, and survey number, while land rights are documented through khatiayan number, land certificate, land deed, etc. [1]. The compilation of essential information about land, along with its ownership details, constitutes a land record (LR) [2].

To manage and organize these sensitive records, many countries employ land record management systems (LRMS). However, there are concerns globally regarding the reliance on outdated paper-based approaches for maintaining these records. Such methods expose the system to higher risks, including data inaccuracies, vulnerabilities to fraud, and challenges in verifying the authenticity of documents. There is a clear and urgent need to modernize these systems, adopting secure digital solutions to enhance data integrity, accessibility, and efficiency, along with mitigating forgery and unlawful land transactions. This transition promises to improve transparency and reliability in land administration on a global scale [2,3].

In recent decades, the shift from traditional paper-based to digital systems has become increasingly prevalent, where governmental offices use centralized dedicated servers to store the LRs of the user. While this transition has streamlined some aspects of record-keeping, it introduces new challenges. The reliance on a single central server creates a single point of failure—if the server shuts down, the entire system could fail. Furthermore, placing full trust in a single authority raises concerns about potential manipulation of records. The traditional centralized systems often store LRs in plain format, raising privacy concerns due to the sensitivity of the information [2].

Additionally, the process of land trading remains laborious and inefficient, especially in Bangladesh. A prospective buyer must first verify the land ownership status at a land office, then meticulously examine and authenticate various documents such as the BIA deed, khatiayans, and mutation records, all in favor of the seller. After a physical inspection of the land, a transfer deed is prepared. Finally, the buyer proceeds to the relevant sub-registry office to submit the registration receipts and associated fees. This comprehensive process can take approximately two months to complete [4].

Some existing studies have shed light on blockchain-based decentralized LRMS to address vulnerabilities in traditional LR systems [2,5–7]. However, the systems often partially tackle the issues such as single-point failures, data manipulation, and limited security, leaving them still susceptible to risks. In response, this paper proposes a blockchain-based LRMS that directly addresses these challenges by enhancing security, efficiency, and reliability. Our approach leverages blockchain technology to reduce single-point failure risks, strengthen data integrity, and streamline the land transaction process. Experimental results demonstrate the system's effectiveness and scalability, establishing it as a promising solution for modern LRMS. The specific contributions of the paper are referred to as follows:

i. This study proposes a blockchain-based LRMS that ensures LR privacy, LR integrity, and facilitates a decentralized land trading platform.
ii. This paper develops a new hybrid chaotic system (HCS) consisting of the tent map and logistic map, facilitating a wide chaotic range that increases the randomness of the system, the symmetric key, etc.
iii. This work ensures LR privacy by introducing a new lightweight cryptosystem that requires two secret keys for encryption and decryption operations. The secret keys are generated using the HCS, which ensures the robustness of the keys.
iv. This study preserves LR integrity by employing both InterPlanetary File System (IPFS) and blockchain technology, which also makes the LRMS robust, decentralized, and scalable.
v. This study introduces a decentralized land advertising platform that accelerates the process of ownership change. A seller (land-owner who wants to sell land) posts on the platform, whereas a buyer (a person who wants to buy) searches for the expected land using the platform.
vi. The work proposes a new technique of mapping character to integer called the dynamic character to integer (C2I) table using the proposed HCS. Employment of the dynamic C2I table increases the level of security and lessens around 33% overhead in the conversion of text data to integer data compared to the ASCII table.

The rest of the paper is organized as follows: Section 2 delves into a comprehensive analysis of the related works, while Section3 explains the preliminaries. Section 4 describes the contribution of this work, and Section 5 demonstrates the proposed LRMS. Section 6 provides a rigorous security and privacy analysis, while Section 7 analyzes the complexity of the system. Experimental results are discussed in Section 8, and Section 9 concludes the paper.

2. Related work

Numerous prior works have already been presented to tackle the constraints associated with traditional LRMS. Table 1 provides a summary of these different approaches.

The system described in Ref. [5] comprised three stages and utilized the Ethereum and hybrid platform to create a prototype. Its main features included data synchronization, transparency, and simplified access to land title information in Bangladesh. The system emphasized user-centric control through decentralized identifiers [8] and public key infrastructure [9] management. Smart contracts were designed in detail to facilitate the functioning of the system. On the other hand, the system presented in Ref. [6] was tailored for land registration in Pakistan. It employed a private blockchain to ensure trust and transparency within the system. However, unlike the system in Ref. [5], it only provided a conceptual framework without implementing a functional prototype. One similarity between the systems was that they stored plain land registry data directly on the blockchain.

The system developed in Ref. [7] presented a smart contract targeting specific cases, including ownership transfer, ownership share, real estate split or merge, etc. It is implemented in Serbia, addressing one of the major concerns, like double-spending prevention. However, the study limited the possibility of trading real estate. Another framework introduced in Ref. [10] is also based on decentralized technology, facilitating smart contracts. Similar to Refs. [5] and [6], both Refs. [7] and [10] store plain LR on the blockchain.

Ensuring the security of land transactions, a decentralized land registration scheme was designed [11]. By utilizing SHA-256 for hashing and elliptic curve cryptography for signatures, the system ensured that transactions were securely recorded on the blockchain, protecting against tampering and unauthorized access. These cryptographic techniques play a crucial role in the overall security and trustworthiness of the system. The scheme underwent evaluation utilizing a proof of work (PoW) consensus mechanism, encompassing a network of 12 nodes and a limited scope of 200 transactions. It’s worth noting that PoW can potentially incur substantial resource consumption.

The scheme implemented in Ref. [12] employed smart contracts that facilitated authentic and conclusive rights on the ownership land titling system. It provided a mechanism for purchasers to ensure that the land they intended to buy was indeed the correct plot and that the seller’s ownership was unambiguous, while lessening the disputes and forgery. However, it is important to note that the study did not take privacy concerns related to land data into account.

Focusing on a quicker transaction process, the system in Ref. [13] presented a smart contract application in the field of land administration employing user-controlled identities. Besides, the scheme eliminated one of the major problems in blockchain technology called double-spending. However, the system did not include any data encryption mechanism to protect the privacy of land information.

The work [14] developed a blockchain-based framework for digitizing property transactions, focusing on the risk of document forgery and other fraudulent activities in India. They incorporated IPFS, which is a peer-to-peer (P2P) swarm network, along with different regional registry offices into their framework. Their proposed system reduced transmission overhead for multicasting nodes and message exchange communication overhead, mostly compared to existing systems, while the LR was stored in a plain format. Another similar work by the authors [15] also stores plain LR on the blockchain that potentially reveals the privacy of the land owner, LR, etc.

The system developed in Ref. [16] employed two types of blockchains such as a main chain for storing publicly accessible metadata and a sidechain for housing non-transactional data like images, contracts, PDFs, and related information. The primary focus was on leveraging blockchain technology to enhance land registry data storage, streamline information management, accelerate authentication, and optimize storage efficiency for more efficient LR searches within the blockchain. The proposed approach incorporates a summary file and employs IPFS within a P2P network. The summary file concept is utilized to facilitate document searches within the sidechain, allowing queries based on registry attributes and restricting access to authorized parties involved in the respective transaction. This system also brings to light the potential privacy concerns associated with LRs, which could pose significant risks.

Another LRMS developed in Ref. [2] considered privacy, integrity of LR, including a land trading platform. However, the system has several notable concerns. Firstly, the system exploited the ElGamal asymmetric cryptosystem to enhance the LR privacy, resulting in cipher data that is approximately 6–24 times larger than the plain data. Also, the public key cryptosystem requires a high processing time, which is challenging in this domain. Secondly, they stored the cipher LR on the blockchain, which may make the system impractical as large data stored on the blockchain make the system surprisingly slower. Thirdly, while the system facilitated land trading, it lacked explicit details about the platform's mode of operation. The BACP-LRS proposed in Ref. [17] ensured privacy of LR using the AES technique along with a 128-bit key and integrity through blockchain and IPFS integration, but faces limitations. It used RSA encryption to secure the symmetric key, potentially impacting performance during key management.

The work in Ref. [18] developed a mathematical model for a blockchain-based land registration system that employed different mathematical models for LR encryption, transaction processing time, cost reduction, user adoption, etc. The reduced transaction processing time, along with cost reduction, made their system scalable. However, there is no specific encryption technique in their work. Another scheme in Ref. [19] proposed a scalable LRMS incorporating a P2P swarm named IPFS that employed trusted LRD (land registration department). Besides, the authors also proposed a consensus algorithm based on the trust value of the miners that makes the system scalable by reducing the computation time. Compared to PoW, the scheme achieved 58.94% less message exchange per consensus, along with 24.44% less time than the existing load-balanced method.

Table 1 Comparison among different land record management systems.

System	Security features	Scalability	Adopted country	Self-sovereign identity principles	Blockchain model
Ref. [5]	Immutability, transparency	Employed three phases supporting broader range of adoption	Bangladesh	User-centric control, Difference-in-Differences (DID) management & public key infrastructure (PKI) usage	Permissionless, hybrid
Ref. [6]	Immutability	–	Pakistan	Authorized user control, direct interaction	Permissionless, permissioned

System	Security features	Scalability	Adopted country	Self-sovereign identity principles	Blockchain model
Ref. [7]	Immutability, prevention of double spending	–	Serbia	Digital verification by participants	Permissionless
Ref. [10]	Immutability, authentication scheme	–	India	Role-based access control	Permissioned
Ref. [11]	Immutability, privacy through SHA-256 & ECC	–	India	User-controlled identity	–
Ref. [12]	Immutability	Incorporated multiple channels	India	Usage of PKI	Permissioned
Ref. [13]	Immutability, transparency	–	Serbia	User-controlled identity	–
Ref. [14]	Immutability	Optimized consensus algorithm that reduces transaction time by ~50%	India	User-controlled identity	Permissioned
Ref. [15]	Immutability	Developed a trust-value-based consensus algorithm that reduces overhead transmissions by ~50%	–	User-controlled identity	–

System	Security features	Scalability	Adopted country	Self-sovereign identity principles	Blockchain model
Ref. [16]	Immutability, secure data exchange using the two-way peg protocol	Used sidechain to offload the mainchain	India	User-centric identity management	–
Ref. [2]	Immutability, privacy through the ElGamal cryptosystem	Designed a mapping table to reduce transaction length and time	Bangladesh	User-controlled identity	Permissionless
Ref. [17]	Immutability, privacy via AES	Employed a multi-layered architecture	–	User-controlled identity	Permissionless
Ref. [18]	Immutability, privacy via an encryption technique	Ability to handle a growing volume as the user base expands	–	User-controlled identity	–
Ref. [19]	Immutability	Developed a trusted node consensus algorithm, message exchange is 58.94% less than traditional PoW	India	User-centric identity management	Permissioned
Ref. [20]	Immutability	–	Pakistan	National authority for verifying identities	Permissioned

System	Security features	Scalability	Adopted country	Self-sovereign identity principles	Blockchain model
Ref. [21]	Immutability	–	Pakistan	National authority for verifying identities	Permissioned
Ref. [22]	Immutability, privacy through encapsulating the security payload, solves routing attacks	–	Australia	Centralized identity management	Permissioned
Ref. [23]	Immutability, transparency	–	India	Users control identity	Permissionless

The work in Ref. [20] proposed a blockchain-based LRMS employing a smart contract. The LRD of their system stored LR both locally and on the Blockchain network, which made the impractical. Every node in the network may access the LR from the ledger of LRD. Another similar work in Ref. [21] also gave the full authorization to the LRD to maintain the LR of the user. Hence, the system also raises practicality concerns. In contrast, to prevent the LR from routing attacks, the work in Ref. [22] proposed a VPN-based land title transfer system implemented in Hyperledger Fabric. The system employed the encapsulating security payload protocol to encrypt the LR. Another system in Ref. [23] used smart contracts for the Ethereum blockchain for the system implementation, allowing users to control identities. However, unlike Ref. [22], the system did not incorporate any encryption technique. Moreover, both Refs. [22] and [23] did not consider other major concerns like distributed LRD, scalability, etc.

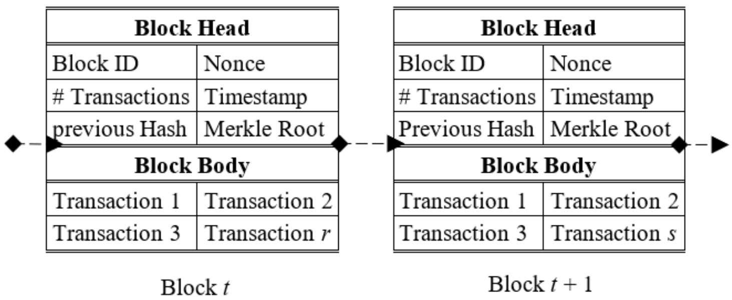


Fig. 1. Blockchain simplified structure.

After analyzing the existing state-of-the-art works, certain research gaps have been identified that must be addressed in a realistic LRMS. Firstly, there is a crucial concern related to LR privacy. Some previous works stored plain LR directly on the blockchain, potentially compromising privacy. While a few attempts were made to enhance LR privacy using asymmetric cryptosystems, they proved to be impractical due to high data overhead and time consumption. Besides, none of the systems examined embedded a decentralized land advertising platform to facilitate the advertising and selling of land, providing a seamless experience for potential buyers. Additionally, some of the existing systems suffered from significant time delays in processing transactions and creating new blocks, rendering the LRMS impractical in real-world scenarios. However, the proposed LRMS in this paper addresses these limitations, striving to create a more comprehensive and effective solution that ensures LR privacy, incorporates a decentralized land advertising platform, and optimizes transaction processing for practical use.

3. Preliminaries

3.1. Blockchain

Blockchain is a decentralized, distributed, and immutable technology that stores historical records of all transactions that happen in a P2P network. Nowadays, it has found widespread application across diverse sectors, encompassing e-governance, supply chain management, public administration, healthcare, etc. [24,25]. Data are stored in blocks that are cryptographically linked to create a chain. The hash pointer connecting each block to the previous one ensures the immutability of the blockchain. A simplified blockchain structure is illustrated in the following Fig. 1.

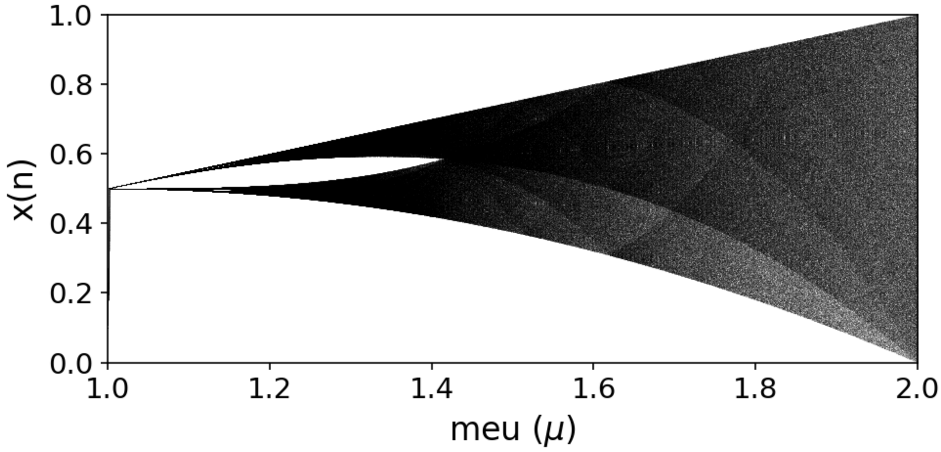
Each block consists of two portions: (i) the block header and (ii) the block body. The header portion contains essential metadata such as block ID, previous block hash, total number of transactions, nonce, Merkle root, timestamp, etc. On the other hand, the body portion houses the transaction data itself. The Merkle root is derived from a binary hash tree known as the Merkle tree. Secure hashing algorithm (SHA)-256 is a widely utilized hashing algorithm in this domain [26].

3.2. IPFS

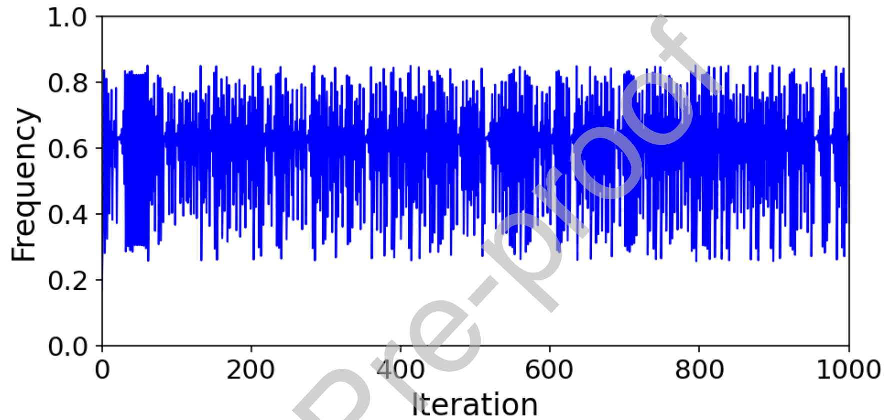
The IPFS is a protocol and network established in 2015 with the aim of establishing a P2P approach to storing and sharing files within a distributed file system [27]. It offers a decentralized approach to content distribution by creating a network of nodes that collectively store and serve content. The aim is to overcome the limitations of the conventional centralized client-server architecture of the web, which can lead to issues such as single points of failure, censorship, etc. Each piece of content is identified by a unique hash, and these hashes are used to locate and retrieve the content from the network.

3.3. Tent Chaotic Map

The tent map is a piecewise-linear map that exhibits folding and stretching behaviour. It is defined by a parameter that determines the slope of the linear segments. The tent map can generate chaotic sequences that are uniformly distributed, represented using Eq. (1). In the tent map, μ is the system parameter in the range of [0, 2].



(a) Bifurcation diagram of Tent map.



(b) Histogram analysis of Tent map for 1000 iterations.
(μ = 1.7 and x_n = 0.1)

Fig. 2. Tent chaotic map.

$$x_{n+1} = \begin{cases} \mu \times x_n & 0 \leq x_n \leq \frac{1}{2} \\ \mu \times (1 - x_n) & 0 < x_n \leq 1 \end{cases} \quad (1)$$

The most important property of a chaotic map is the bifurcation diagram that shows the behavior of a dynamic system by varying the system parameter μ. The bifurcation diagram of the Tent chaotic map is represented in Fig. 2a, where Fig. 2b shows the histogram analysis of the map of 1000 values/iterations when μ = 1.7 and x_n = 0.1. One of the basic limitations of the bifurcation diagram is a small chaotic range where the values exist between nearly 0.3–0.8, as illustrated in Fig. 2.

3.4. Logistic Chaotic Map

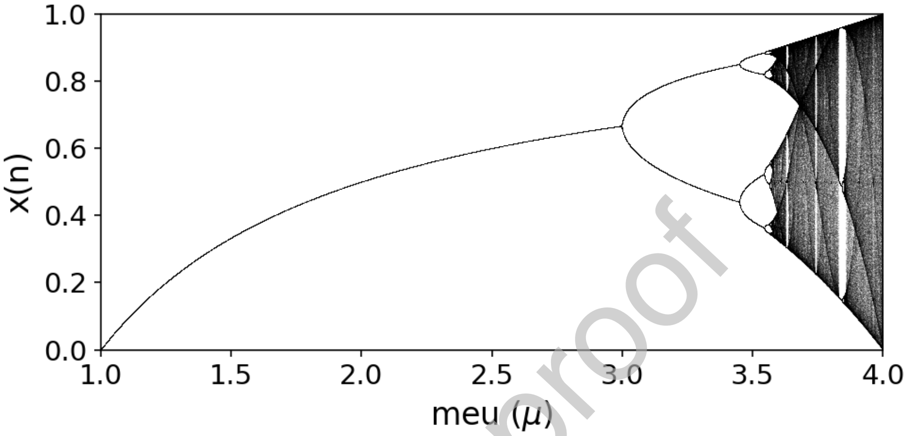
A logistic map is a nonlinear map defined by a parameter that controls the behaviour of the map. The logistic map can produce sequences that exhibit sensitive dependence on initial conditions and parameter values. The map is represented using Eq. (2). In the logistic map, the system parameter μ is in the range of [0–4].

$$x_{n+1} = \mu \times x_n \times (1 - x_n) \; (2)$$

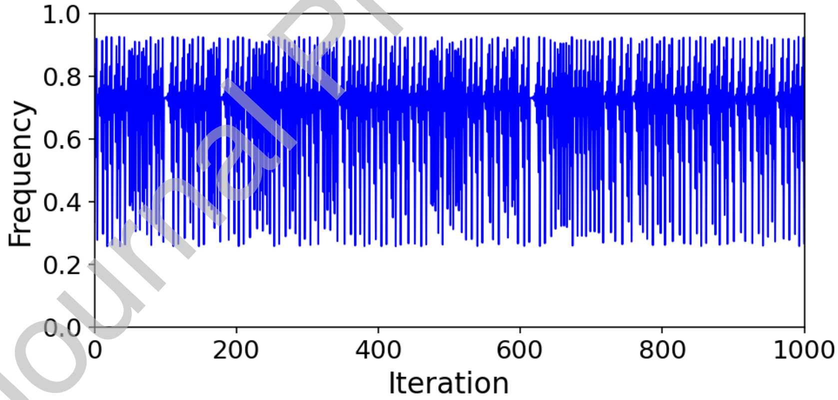
The bifurcation diagram of the Tent chaotic map is represented in Fig. 3, where Fig. 3a illustrates the bifurcation diagram and Fig. 3b shows the histogram analysis, respectively, of the map of 1000 values/iterations when $\mu = 3.7$ and $x_n = 0.1$. The bifurcation diagram of the logistic map also generates a small chaotic range that exists between nearly 0.3–0.9, which is the weakness of the map as shown in Fig. 4.

3.5. *C2I table*

A character-to-integer mapping scheme named the C2I table is proposed in Ref. [2]. The goal was to reduce transformation overhead in comparison to the ASCII table.



(a) Bifurcation diagram of Logistic map.



(b) Histogram analysis of Logistic map for 1000 iterations.
($\mu = 3.7$ and $x_n = 0.1$)

Fig. 3. Logistic chaotic map.

By considering the useful 95 individual characters, the C2I requires only 2 decimal digits to convert a single character to its corresponding integer, while the ASCII table requires 3 decimal digits. This optimization is possible because many ASCII characters are irrelevant in textual representation. This clearly demonstrates that when employing the C2I table for converting text data to integer data, there is a notable reduction of approximately 33% in overhead compared to using the ASCII table. A sample C2I table is shown in Table 2. However, it is important to note that the C2I table has a limitation in its static nature, which could potentially make it predictable and pose a security risk.

4. Contribution of the work

This section describes the contribution of this paper. First, it demonstrates a hybrid chaotic map, combining the tent and logistic maps, statistical test, and key generation using it. Then, the dynamic character-to-integer mapping scheme is described. After that, the developed symmetric encryption technique is demonstrated. Finally, the decentralized land advertising platform is illustrated.

Table 2 Character to integer (C2I) mapping.

Character	Integer	Character	Integer	Character	Integer	Character	Integer	Character	Integer	Character	Integer
0	00	H	17	Y	34	p	51	‘	68	^	85
1	01	I	18	Z	35	q	52	(	69	_	86
2	02	J	19	a	36	r	53	)	70	[	87
3	03	K	20	b	37	s	54	□	71	\	88
4	04	L	21	c	38	t	55	+	72	]	89
5	05	M	22	d	39	u	56	,	73	‘	90
6	06	N	23	e	40	v	57	-	74	~	91
7	07	O	24	f	41	w	58	.	75	{	92
8	08	P	25	g	42	x	59	/	76	|	93
9	09	Q	26	h	43	y	60	:	77	}	94
A	10	R	27	i	44	z	61	;	78	–	–
B	11	S	28	j	45		62	<	79	–	–
C	12	T	29	k	46	!	63	=	80	–	–
D	13	U	30	l	47	“	64	>	81	–	–

Character	Integer	Character	Integer	Character	Integer	Character	Integer	Character	Integer	Character	Integer
E	14	V	31	m	48	#	65	?	82	–	–
F	15	W	32	n	49	%	66	@	83	–	–
G	16	X	33	o	50	&	67	$	84	–	–

4.1. Hybrid chaotic map

To overcome the limitation of a small chaotic range, this paper proposes a hybrid chaotic map consisting of the tent map and the logistic map, which is used to incorporate the randomness of the C2I table. Benefitting from the unique properties of both the tent and logistic maps, the hybrid scheme is designed. In the hybrid chaotic map, the tent map can introduce randomness to the sequence, while the logistic map can add complexity and nonlinear behaviour. The formula of the hybrid map is represented in Eq. (3). Here, the value of the system parameter μ is in the range of [0, 4]. The following Algorithm 1 shows the proposed new HCS.

$$x_{n+1} = \mu \times x_n \times (1 - x_n) + \frac{(4-\mu)}{2} \times \begin{cases} x_n & ; 0 \leq x_n \leq \frac{1}{2} \\ (1 - x_n) & ; \frac{1}{2} < x_n \leq 1 \end{cases} \quad (3)$$

Algorithm 1: Hybrid chaotic system (HCS)

1. Input: x_n, μ, n

2. Output: $A[\ldots]$

3. Initialize x_n, μ

4. Initialize $A = []$

5. For each i **in** *range* (n) **do:**

6. Calculate: $val_1 = \mu \times x_n \times (1 - x_n)$

7. Calculate: $val_2 = (4 - \mu)/2$

8. Find minimum: $val_3 = min\ (x_n, 1 - x_n)$

9. Calculate: $x_{n+1} = val_1 + val_2 \times val_3$

10. $A.append\ (x_{n+1})$

11. End For

12. Return $A[\ldots]$

The bifurcation diagram of the developed hybrid chaotic map is represented in Fig. 4(a) where Fig. 4(b) shows the histogram analysis of the map of 1000 values/iterations when μ = 3.7 and x_n = 0.1. The HCS generates a very wider chaotic range which indicates the high randomness behavior as shown.

Lemma 1*: Hybrid chaotic map produces:* $x_{n+1} \epsilon (0,1) \rightarrow x_n \epsilon (0,1)$ *where* $\forall\mu[0,4]$.

Proof:

If μ = 0, the HCS reduces to the chaotic tent map.

If μ = 4, the HCS reduces to the chaotic logistic map.

If 0<μ<4 and $0 \leq x \leq \frac{1}{2}$, the HCS produces $x_{n+1} < 1$.

If 0<μ<4 and $\frac{1}{2} < x \leq 1$, the HCS produces $x_{n+1} < 1$.

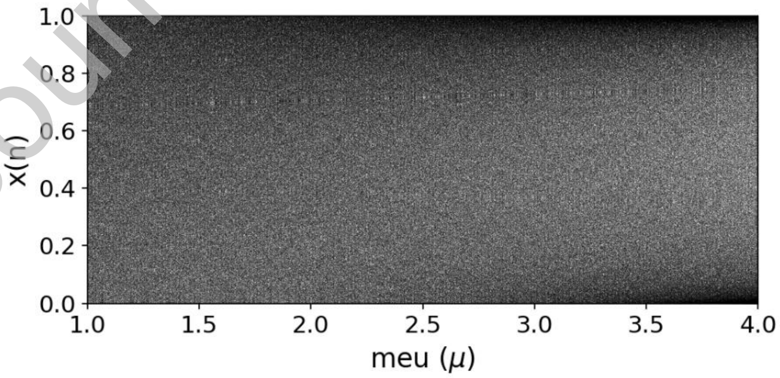
Lemma 2: *Hybrid Chaotic Map facilitates improved randomness and security*.

Proof:

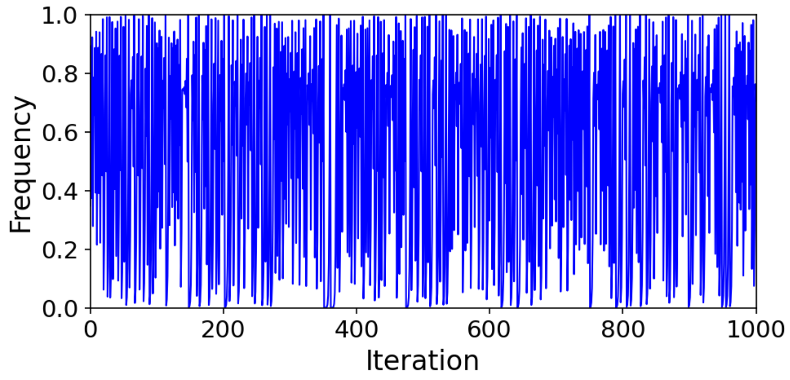
A statistical test suite has been developed by National Institute of Standards and Technology (NIST) to verify the randomness and security of the generated cipher data of a system [28]. This suite comprises 15 different tests, each aimed at evaluating specific aspects of randomness. The input of the NIST test is a binary sequence produced by the HCS. To assess the performance of each test, it estimates the value of *P* where the value >0.01 is considered as a pass. Table 3 presents the statistical analysis of the proposed HCS. The table shows that the chaotic system passes all the statistical tests.

4.2. Key generation using a hybrid chaotic map

This sub-section demonstrates the algorithm to generate a key using the hybrid chaotic map. The algorithm includes four steps and is shown in Algorithm 2.



(a) Bifurcation diagram of Hybrid map.



(b) Histogram analysis of the hybrid map for 1000 iterations.
(μ = 3.7 and x_n = 0.1)

Fig. 4. Proposed Hybrid chaotic map.

Table 3 Nist randomness test analysis for hybrid chaotic system.

Sl. No.	Test name	*P* value	Result
1	Frequency (Monobit)	0.676	Pass
2	Frequency (Block)	0.271	Pass
3	Run	0.188	Pass
4	Longest run	0.277	Pass
5	Binary matrix rank	0.693	Pass
6	Discrete Fourier transform (spectral)	0.852	Pass
7	Non-overlapping template matching	0.611	Pass
8	Overlapping template matching	0.021	Pass
9	Universal	0.334	Pass
10	Linear complexity	0.020	Pass

Sl. No.	Test name		P value	Result
11	Serial cumulative sums	Value 1	0.723	Pass
		Value 2	0.554	Pass
12	Approximate entropy		0.445	Pass
13	Cumulative sums	Forward	0.807	Pass
		Reverse	0.528	Pass
14	Random excursions (state: -4)		0.859	Pass
15	Random excursions variant (state: -4)		0.735	Pass

Step 1: Choose a suitable system parameter μ for the hybrid chaotic map, initial value of seed x_n, desired symmetric key length k_{len}, and number of iterations n. Typically, $0 < \mu < 2$, $0 < x_n < 1$, $128 < k_{\text{len}} < 1024$ and $20 \times k_{\text{len}} < n < \infty$.

Step 2: Generate the pseudo-random numbers using the proposed hybrid chaotic map.

Step 3: Convert the numbers into their corresponding bits.

Step 4: Concatenate the sequence of bits to form a single bit string. Discard bits from left of the bit if the string length is greater than the defined length.

Algorithm 2: Key generation using HCS (KG)

1. Input: x_n, μ, n, k_{len}

2. Output: Key *Key*

3. Initialize x_n, μ, n, k_{len}

4. *Key* = Null

5. Initialize *A* = []

6. *A*[…] = *HCS* (x_n, μ, n) [**Algorithm 1**]

7. For each *val* **in** *A*[…] **do**:

8. Calculate: *val* = *toInteger* (*val*)

9. Calculate: *val* = *toBinary* (*val*)

10. Key += val

11. End For

12. If ($len(Key) > k_{len}$) **do**:

13. Discard ($len(Key) - k_{len}$) bits from left the Key_s

14. End If

15. Return *Key*

4.3. Dynamic C2I table

To overcome the limitation behind the C2I table proposed in Ref. [2], a new dynamic C2I table based on the proposed HCS is developed. The robustness of the table can be boosted by ensuring the dynamic behavior of the table. The design principles of the dynamic C2I table are as follows:

Chaotic dynamism: The root of the dynamic C2I table lies in the HCS, which produces a sequence of pseudo-random numbers based on initial parameters (x_n, μ, n). Even small changes in the parameters lead to a vastly different mapping.

Uniqueness and non-repetition: The intrinsic unpredictability of HCS confirms that the resulting mappings are unique by preventing collisions and repetition.

Scalability and flexibility: The following dynamic C2I table generation algorithm can be adapted to different character sets by adjusting the modulo operation and character list, which make the system more flexible and scalable.

The generation of the proposed dynamic C2I table is presented in the Algorithm 3.

Algorithm 3: Generation of dynamic C2I table (GC2I)

1. Input: x_n, μ, *cnt*

2. Output: C2I table

3. Initialize x_n, μ, n

4. Set *cnt* = 0

5. Initialize A = []

6. Specify the required 95 characters

7. While (*True*) **do**:

8. $A[\ldots] = HCS$ (x_n, μ, n) [**Algorithm 1**]

9. For each x **in** *range* (n) **do**:

10. Calculate: $val = toInteger$ (x)% 95

11. If (*val* not in C2I table) **do**:

12. Assign *val* to a new unassigned character randomly

13. *cnt*+ + **End If**

14. If (*cnt* == 95) **do**:

15. Return C2I table

16. End If

17. End For

18. Update (x_n, μ, n)

19. End while

20. Return C2I table

To implement Algorithm 3, the core steps are: HCS initialization, character set definition, mapping process, and parameter update (if necessary). They are described as follows:

HCS initialization: Initialize the parameters of the HCS to generate the pseudo-random number sequence.

Character set definition: Define the required characters (here 95): digits, uppercase/lowercase letters, and common special symbols.

Mapping process: Convert each chaotic number to an integer and apply the modulo operation (here using 95) to ensure the value falls within the desired range.

Parameter update: Check if all characters in the defined character set have been obtained mapping or not; if not, update HCS's parameters to map the unassigned characters.

After each session, update the initial parameters of the chaotic system to regenerate a new dynamic table.

4.4. Proposed lightweight cryptosystem

Table 4 Operation.

Sl. No.	Code	Operation		
		Operation1	Operation2	Swap
1	00	chunk #1 $\oplus$ key_d	chunk #2 $\oplus$ key_d	No
2	01	chunk #1 $\oplus$ key_d	chunk #2 $\oplus$ key_s	Yes
3	10	chunk #1 $\oplus$ key_s	chunk #2 $\oplus$ key_d	Yes
4	11	chunk #1 $\oplus$ key_s	chunk #2 $\oplus$ key_s	Yes

This paper demonstrates a new symmetric encryption scheme. It incorporates swap and XOR operations for encryption purposes. The operation is selected from Table 4. Note that Table 4 shows only one coding scheme, while there exist 23 more coding patterns. The developed technique requires two keys, i.e., the determinant key and the symmetric key. The design principles of the proposed lightweight cryptosystem are as follows:

Low computational overhead: The encryption and decryption stages of the proposed lightweight cryptosystem are designed based on the hybrid chaotic map, swap operation, and XOR operation, which makes the cryptosystem suitable for systems with limited computational power.

High key security: The usage of an HCS consisting of the tent map and the logistic map to generate a determinant key and a symmetric key ensures strong randomness and resilience against attacks.

Unpredictability mapping: The employment of a dynamic C2I table introduces variability in the character-to-integer conversions.

Randomized operation: The random usage of the determinant key and the symmetric key for the swap and XOR operations enhances data confusion and security through random transformations.

The encryption and decryption processes are described as follows:

4.4.1. Encryption process

The working procedure of the encryption process is described with the following steps and illustrated in Algorithm 4.

Step 1: Read data to be encrypted.

Step 2: Generate a dynamic C2I table.

Step 3: Convert each character into its corresponding integer value based on the C2I table.

Step 4: Make equal-length chunks of the sequence of integer values and convert them into corresponding bits. Apply 0-padding if necessary.

Step 5: Set variable m and variable r to 1.

Step 6: Repeat Step 6 to Step 10 until unencrypted chunks are available (i.e., $r + 1$ is the last chunk).

Step 7: Scan m^{th} and $(256 - m)^{th}$ bit of the determinant key key_d.

Step 8: Select operations based on two bits from the following Table 4.

Step 9: Perform the selected operations to r^{th} and/or $(r + 1)^{th}$ chunk using the symmetric key key_s or determinant key key_d. Also, swap positions of the chunks (if necessary).

Step 10: Step 10: Increase the value of m and r.

Step 11: Merge the binary chunks to form a single encrypted chunk.

Step 12: Convert each 6-bits into its corresponding integer value.

Step 13: Finally, replace each integer value with its corresponding character using the C2I table.

Algorithm 4: Proposed encryption process

1. Input: key_d, key_s, n, k_{len}

2. Output: Cipher data C_D

***Stage* 1: Parameter Initialization**

3. Initialize x_{n1}, μ_1, n_1, k_{len1}

4. $Key_d = KG\ (x_{n1}, \mu_1, n_1, k_{len1})$ [**Algorithm 2**]

5. Initialize x_{n2}, μ_2, n_2, k_{len2}

6. $Key_s = KG\ (x_{n2}, \mu_2, n_2, k_{len2})$ [**Algorithm 2**]

7. Initialize x_{n3}, μ_3, cnt

8. Generate C2I table: $GC2I\ (x_{n3}, \mu_3, cnt)$ [**Algorithm 3**]

9. Read data D

10. Initialize dummy string str

***Stage* 2: Plain Data Conversion** (*string → char → binary*)

11. For each *char* **in** *D* **do**:

12. $int\_val = C2I\ [char]$

13. $str += int\_val$

14. End For

15. $\Omega[1\ldots s]$ = *toBinary*($\Omega[1\ldots s]$)

16. $\Omega[1\ldots s]$ = *makeEqualLengthChunks*(*str*)

***Stage* 3: Operation** (*XOR* and *swap*)

17. $m = 1$

18. For each *r* **in** *range*(*s*) **do**:

19. $b_0 = Key_d\,[m]$

20. $b_1 = Key_d\,[256 - m]$

21. $chnk_1 = \Omega[r]$

22. $chnk_2 = \Omega[r + 1]$

23. Increase: $r = r + 1$ and $m = m + 1$

24. If ($b_0b_1 = 00$) **do**:

25. $chnk_1 = key_d$ XOR $chnk_1$

26. $chnk_2 = key_d$ XOR $chnk_2$

27. End If

28. If ($b_0b_1 = 01$) **do**:

29. $chnk_1 = key_d$ XOR $chnk_1$

30. $chnk_2 = key_s$ XOR $chnk_2$

31. *continue*

32. End If

33. If ($b_0b_1 = 10$) **do**:

34. $chnk_1 = key_s$ XOR $chnk_1$

35. $chnk_2 = key_d$ XOR $chnk_2$

36. End If

37. If ($b_0b_1 = 11$) **do**:

38. $chnk_1 = key_s$ XOR $chnk_1$

39. $chnk_2 = key_s$ XOR $chnk_2$

40. End If

41. Swap chunks: *swap* ($chnk_1$, $chnk_2$)

42. End For

43. $S \leftarrow$ Merge all chunks to form a single cipher chunk

***Stage* 4: Cipher Data Conversion** (*binary* → *char* → *string*)

44. For each 6-*bit* **in** S **do**:

45. $int\_val = toInteger(6\text{-}bit)$

46. $char = C2I\ [int\_val]$

47.C_D + = *char*

48. **End For**

49. Return Cipher data C_D

4.4.2. Decryption process

The working procedure of the decryption process is described with the following steps and also illustrated in Algorithm 5.

Step 1: Read encrypted data.

Step 2: Get the value of m and r, the dynamic C2I table, the determinant key, and the symmetric key.

Step 3: Replace each character with its corresponding integer value.

Step 4: Convert each integer value into its corresponding 6-bit binary value.

Step 5: Make an equal-length chunk of the sequence of bits.

Step 6: Repeat Steps 6–10 until all the chunks get decrypted.

Step 7: Scan m^{th} and $(256 - m)^{th}$ bit of the determinant key key_d.

Step 8: Select operations based on two bits from Table 4.

Step 9: Perform the selected operations to r^{th} and/or $(r+1)^{th}$ chunk using the symmetric key key_s or determinant key key_d. Also, swap positions of the chunks (if necessary).

Step 10: Decrease the value of m and r.

Step 11: Merge all binary chunks to form a single chunk. If necessary, discard 0-padding.

Step 12: Convert into an integer-valued chunk.

Step 13: Convert each 2-digit integer into the corresponding character using the dynamic C2I table and get the plain data.

4.5. Decentralized advertising platform

This paper develops a decentralized land advertising platform to connect the buyers and sellers in a transparent and secure way that makes the trading process easier. However, this decentralized approach uses blockchain technology exclusively for land advertising purposes without storing the transferring data. The mechanism of the platform is described as follows.

Blockchain for ad listings: It leverages blockchain to decentralize the advertising process. By providing relevant details, sellers list their land for sale (e.g., location, price, size). Once the listing is created, it is stored on the blockchain and accessible to all potential buyers.

Seller authentication: Before listing the land, sellers must authenticate themselves through the platform, either via a decentralized identity verification process. This ensures that only legitimate users can post land advertisements.

Verification of listings: While the platform doesn't store land ownership data, it implements a verification process to confirm the authenticity of the land details provided. This could involve cross-referencing external decentralized oracle services or registries that confirm ownership and other details.

Buyer interaction: Buyers who wish to purchase land can log in to the platform, browse or search for suitable properties, and contact the sellers. The decentralized nature of the platform removes the barriers to accessing the available lands and avoiding any manipulation of search results.

Algorithm 5: Proposed decryption process

1. Input: key_d, key_s, n, m, r, k_{len}

2. Output: Plain data D

***Stage* 1: Reading Data**

3. Read m, r and C2I table

4. Read Key_d, Key_s

5. Read cipher data C_D

***Stage* 2: Cipher Data Conversion** (*string* → *char* → *binary*)

6. For each *char* **in** C_D **do**:

7. *int_val* = *C2I* [*char*]

8. *bin_val* = *toBinary*(*int_val*)

9. *str* + = *bin_val*

10. End For

11. Ω[1…*s*] = *makeEqualLengthChunks*(*str*)

***Stage* 3: Operation** (*XOR* and *swap*)

12. $m = 1$

13. While (*True*) **do**:

14. $b_0 = Key_d\,[m]$

15. $b_1 = Key_d\,[256 - m]$

16. $chnk_1 = \Omega[r]$

17. $chnk_2 = \Omega[r + 1]$

18. Decrease: $r = r - 1$ and $m = m - 1$

19. If ($b_0 b_1 = 00$) **do**:

20. $chnk_1 = key_d$ XOR $chnk_1$

21. $chnk_2 = key_d$ XOR $chnk_2$

22. *continue*

23. End If

24. If ($b_0 b_1 = 01$) **do**:

25. $chnk_1 = key_d$ XOR $chnk_1$

26. $chnk_2 = key_s$ XOR $chnk_2$

27. End If

28. If ($b_0 b_1 = 10$) **do**:

29. $chnk_1 = key_s$ XOR $chnk_1$

30. $chnk_2 = key_d$ XOR $chnk_2$

31. End If

32. If ($b_0 b_1 = 11$) **do**:

33. $chnk_1 = key_s$ XOR $chnk_1$

34. $chnk_2 = key_s$ XOR $chnk_2$

35. End If

36. Swap chunks: *swap* ($chnk_1, chnk_2$)

37. End For

***Stage* 4: Data Conversion** (*binary* → *integer* → *char* → *string*)

38. $\Omega[1 \ldots s] = toInteger(\Omega[1 \ldots s])$

39. *str* ← Merge all integer chunks to form a single chunk

40. For each 2-*digit* **in** *str* **do**:

41. *char* = *C2I* [2-*digit*]

42. *D* += *char*

43. End For

44. Return Plain data *D*

The interaction details among seller, buyer, and decentralized land advertising platform are illustrated in section V-F (phase 2).

5. Proposed LRMS

This section demonstrates the architecture and working procedure architecture of the proposed blockchain-based LRMS and illustrates it in Fig. 5.

5.1. System model

Scheme server (SS): An entity responsible for the registration of other entities. It acts to verify the identity of an entity and bind cryptographic keys for the required entity. Both the buyer and seller get digital certificates from the SS.

Buyer (B): An individual seeking to acquire property, referred to as B, selects a piece of land and engages in negotiations with the seller to reach an agreement.

Seller (S): A landowner looking to sell their property.

Decentralized ad agency (DAA): A decentralized advertising platform for land listings. S lists their available land for sale on the platform, while B searches for and browses properties that meet their requirements within this digital marketplace.

Bank (Bn): An institution that plays as an intermediary between B and S for a transaction.

Land registration department (LRD): A government agency responsible for overseeing land-related data management, performing encryption-decryption, employing decentralization technology as storage, etc. The LRD holds some coordinating entity for contact purposes.

IPFS: A distributed and P2P technology designed to create a content-addressable, decentralized method for storing and sharing data. IPFS aims to provide a more robust, permanent, and efficient way of sharing and accessing data across the internet.

Blockchain server (BS): An immutable, decentralized storage technique for storing encrypted data by LRD.

5.2. Trust Model

SS: The SS should be a trusted entity considered in this system, which is logical.

Bn: The Bn should be trusted for transaction purposes here. Obviously, the system can easily incorporate cryptocurrency or a token to make transactions without trusting a third-party Bn.

LRD: This system considers the LRD semi-trusted. The system will fail if all the LRD entities act dishonestly.

5.3. *Overview of the proposed LRMS*

Let a user who wants to sell land advertise it to the *DAA*. It is obvious that the S needs to request the LR from the LRD by showing its block number BN_{BC}. The LRD requests metadata of LR from BS using the IIPFS. After getting the metadata, the LRD requests cipher LR from the IPFS using the index of IPFS. Then the LRD decrypts the cipher LR and sends it to the S. Any interested user who wants to buy the land the user creates a deed and signs it using its private key. Then B sends the deed to the S. If the S agrees, the S also signs the deed and sends it back to the B. Now, the B sends the signed deed to the Bn. Then the Bn performs the transaction and sends proof to the B. This time the B forwards the deeds and the transaction information to the LRD for further processing.

After receiving the deeds and the transaction information, the LRD verifies it. The LRD encrypts the LR using the proposed symmetric encryption technique. Then LRD stores the cipher LR to IPFS, and IPFS returns the index IIPFS. This time, the LRD stores the index to the blockchain storage, and the blockchain returns the block number IIPFS. The major notations used in the proposed LRMS are shown in Table 5.

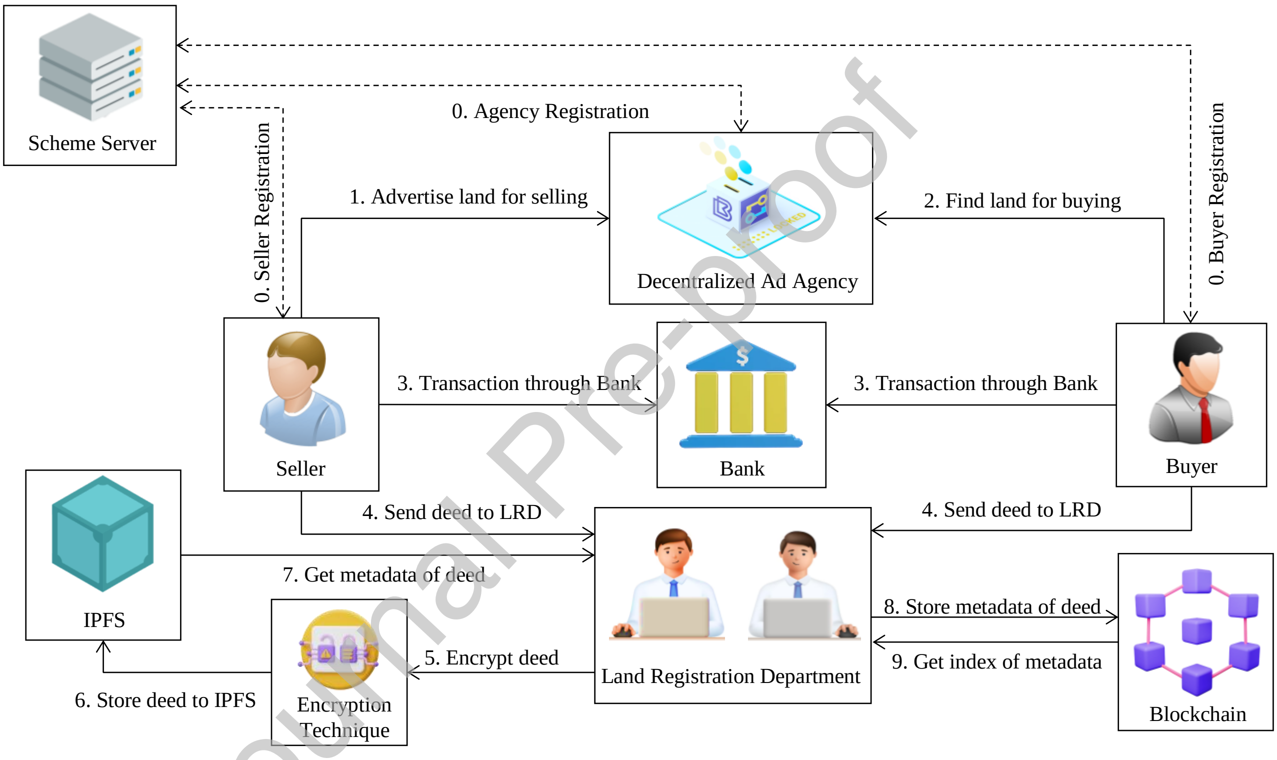


Fig. 5. Working procedure of proposed *LRMS.*

Table 5 Notation used.

Notation	Description
μ	System parameter
B	Buyer

Notation	Description
S	Seller
DAA	Decentralized Ad agency
Bn	Bank
LRD	Land registration department
BS	Blockchain server
BN_{BC}	Block number
$IPFS$	Interplanetary file system
I_{IPFS}	Index of IPFS
Key_d	Determinant key
Key_s	Symmetric key
Key_r	Random key
Key_{dr}	Determinant key ⊕ Random key
Key_{sr}	Symmetric key ⊕ Random key
D	Plain data
C_D	Cipher data
n	Number of iteration
k_{len}	Desired key length

5.4. Formation of LRD and entity selection

As mentioned in the previous section, the LRD consists of several entities. From them, a land owner or buyer selects three entities for managing the LR, where one of the entities will be selected as a coordinator. The following Fig. 6 illustrates the selection process.

5.5. Key distribution inside LRD

The LRD is responsible for both the encryption and decryption processes that incorporate two keys of the land owner, i.e., determinant key key_d and symmetric key key_s. To ensure the secrecy of the scheme, no single entity of LRD stores both keys. Instead, the keys are modified by the land owner or buyer using another random key key_r and distributed all the keys among three entities. This subsection describes the key modification, distribution, and uniting process with the following steps, and Fig. 7 shows the sequence diagram.

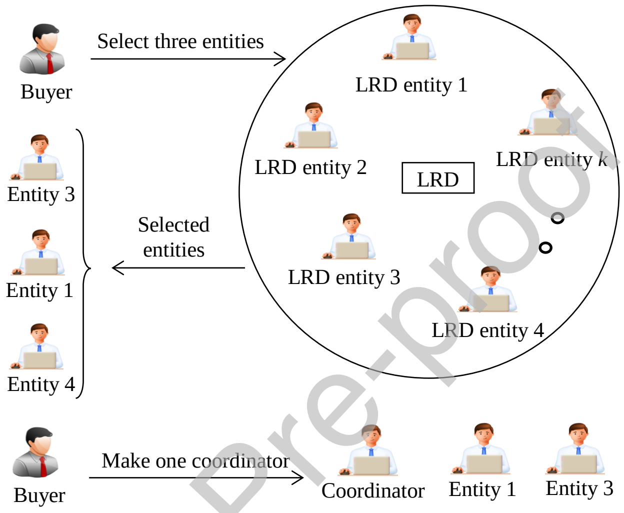


Fig. 6. Entity selection from *LRD*.

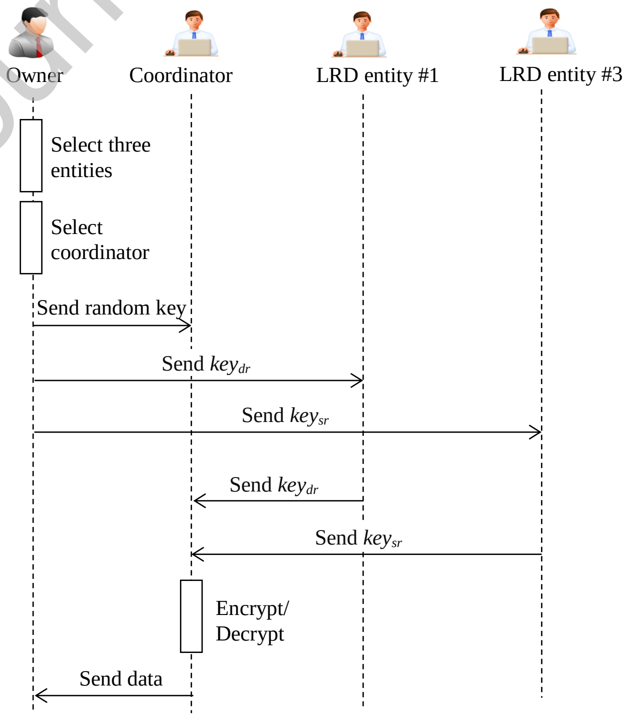


Fig. 7. Key distribution inside LRD.

Step 1: The buyer selects a determinant key_d, a symmetric key_s, and a random key key_r.

Step 2: Then performs two XOR operations, (i) between the determinant key and random key that produces key_{dr} and (ii) between the symmetric key and random key, which produces key_{sr}.

Step 3: Then the buyer sends the random key to the LRD coordinator, key_{dr} to LRD entity #1, and key_{sr} key to LRD entity #2.

Step 4: The coordinator asks the key_{dr} and key_{sr} keys from the LRD entities for both the encryption and decryption processes.

5.6. Working principle of the proposed LRMS

The proposed LRMS works in four phases such as registration, advertisement, transaction, storage, and encryption.

Phase 1: *Registration phase*

At first, every entity needs to be registered on the scheme server. The registration process of each entity is described with the following steps and Fig. 8 illustrates the sequence diagram.

Step 1: An entity or the authorized person of the entity needs to submit an email, national identification (NID), photo, with other necessary information.

Step 2: After performing necessary verifications, the entity gets registered and is provided an ID.

Phase 2: *Advertisement phase*

A user who wants to sell their land advertises it on the decentralized advertising agency. The advertisement process of the seller can be described with the following steps, and Fig. 9 depicts the sequence diagram.

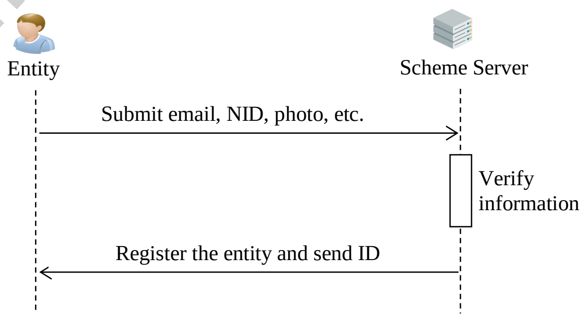


Fig. 8. Registration of an entity.

Step 1: A user who wants to sell their land needs to log in to the decentralized advertising agency.

Step 2: The user then submits the land details to the agency for advertising.

Step 3: The agency verifies the validity of the submitted land information.

Step 4: If verification is OK, then the agency approves the posts. Otherwise, the agency informs the user.

Step 5: Any user who wants to buy land first needs to log in to the decentralized advertising agency.

Step 6: Then he searches for suitable land.

Phase 3: *Transaction phase*

The transaction process between the seller and buyer is described with the following steps, and the time sequence diagram is shown in Fig. 10.

Step 1: After finding the suitable land, the user creates a deed with the necessary information about the land and signs it. Then he sends the deed to the seller.

Step 2: If the seller agrees, he also signs the deed and sends it back to the buyer.

Step 3: At this point, the buyer sends the signed deed to the bank. Then the bank performs the transaction and sends proof to the buyer.

Step 4: This time, the buyer forwards the deeds and the transaction information to the LRD for further processing.

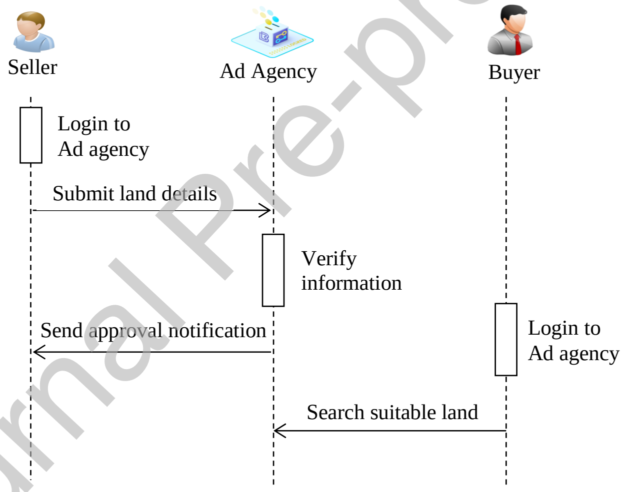


Fig. 9. Advertisement of land information.

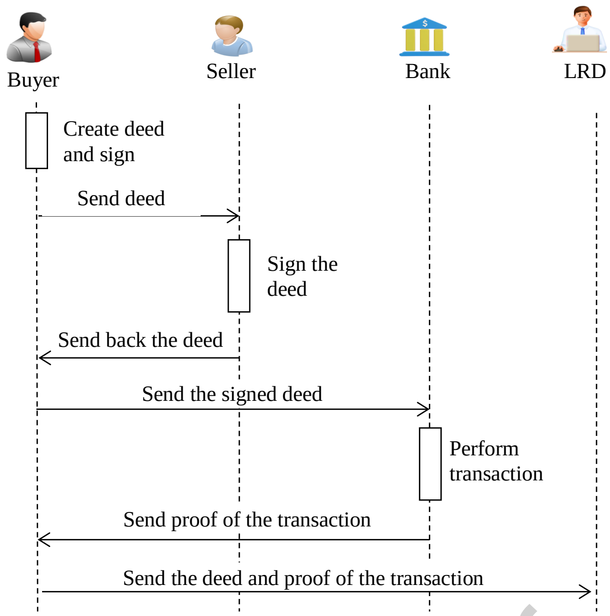


Fig. 10. Transaction between buyer and seller.

Phase 4: *Encryption and storage phase*

This subsection includes the LRD, IPFS, and blockchain server. The storage process of the LRD can be described with the following steps, and Fig. 11 shows the sequence diagram.

Step 1: After receiving the deeds and the transaction information, the LRD verifies it.

Step 2: The LRD encrypts the data using the proposed symmetric encryption technique.

Step 3: Then LRD stores the cipher deeds and transaction information to IPFS, and IPFS returns the index.

Step 4: This time, the LRD stores the index to the blockchain storage, and the blockchain returns the block number.

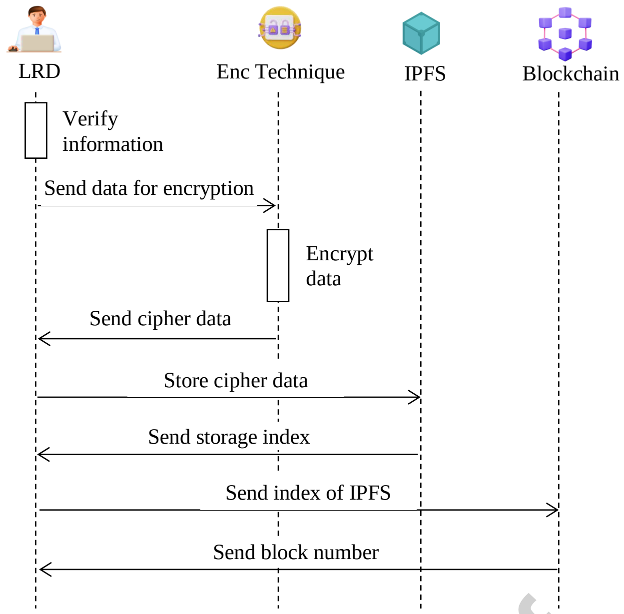


Fig. 11. Storage and encryption of land record.

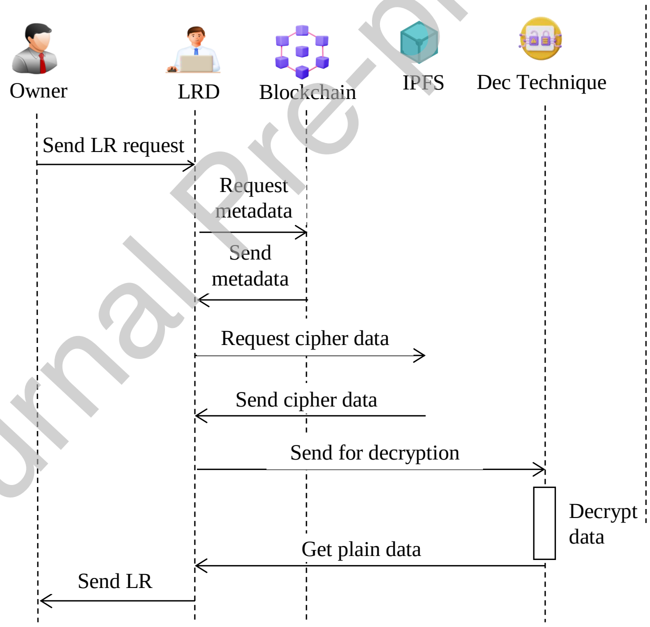


Fig. 12. Query and decryption of cipher land record.

Phase 5: *Query and decryption phase*

This subsection describes the LR query and decryption processes. A user who wants to access its LR sends an access request to the LRD coordinator. Receiving the request, the LRD gets the plain LR using the decryption process and replies to the land owner. The phase is described with the following steps, and Fig. 12 illustrates the sequence diagram.

Step 1: A land owner who wants to access its own LR sends an access request to the LRD.

Step 2: After receiving the access request, the LRD finds the corresponding block number of the owner. Then the LRD gets the metadata of the cipher LR from the blockchain server. This time, the LRD gets the cipher LR, providing the storage index to the IPFS.

Step 3: Then, LRD decrypts the cipher LR using the decryption process of the proposed technique.

Step 4: This time, the LRD sends the LR to the land owner.

5.7. Self-sovereign identity (SSI) of LRMS

The proposed LRMS integrates SSI principles [29] to achieve enhanced security, transparency, and user empowerment. The SSI principles are described as follows:

User control: In the proposed LRMS, users independently manage their cryptographic keys and authorize specific entities (e.g., LRD, banks) to access their data, reducing dependency on intermediaries.

Verifiable credentials: The proposed system can issue verifiable credentials during the Registration Phase, allowing buyers and sellers to prove their identity and ownership seamlessly during transactions. The DAA, banks, and LRD can validate these credentials without requiring sensitive user data, ensuring privacy and efficiency.

However, with SSI, the users maintain full control over their data and share only what is necessary for specific actions. This expands the reach and usability of the LRMS across jurisdictions.

5.8. Real-world regulatory compliance for Bangladesh

The proposed LRMS aligns with the legal framework of Bangladesh by integrating key processes such as Record of Rights validation through the Land Office, mutation application tracking, Revised Survey inspection, obtaining Non-Encumbrance Certificates from the Sub-Registry Office, etc., through the LRD. Then the buyer automates stamp duty and tax calculations, and initiates secure payments. Digital deed creation and signing by buyers and sellers ensure legal compliance, while registration applications are streamlined through the Sub-Registry and Land Revenue Offices. Finally, ownership updates are securely stored on blockchain and IPFS by the LRD, enhancing transparency and trust in property transactions. However, the other countries can fit their regulatory compliance and legal frameworks into the proposed LRMS [30].

6. Security and privacy analysis

The security of the proposed LRMS depends on the security of the newly developed lightweight cryptosystem, the HCS, and the dynamic C2I table. This section provides a detailed security analysis of the proposed scheme.

Corollary 1: With the extensive keyspace, the system ensures the high privacy of LR.

Proof:

The keyspace, which encompasses the entire set of possible keys applicable within a system, is a crucial determinant of security. A larger keyspace means there are more possible keys, which makes it harder for an attacker to guess the correct key. The strength of an encryption technique is heavily dependent on the size and randomness of its keyspace. The keyspace of the proposed technique is calculated as shown in Table 6. In modern cryptography, at least a 128-bit key is recommended for secure systems, while keys of 256 bits or more are commonly used for systems that require higher levels of security. The keyspace size of the proposed encryption technique is 2^{396}, which ensures the high privacy of LR.

Table 6 Keyspace calculation.

Keys	Bits	Space
Determinant key key_d	256	2^{256}
Symmetric key key_s	128	2^{128}
C2I mapping scheme	7	2^7
Operation coding	5	2^5
Total	396	2^{396}

Corollary 2: The LRMS resists brute-force attack with robust key complexity and additional security.

Proof:

In this attack, the attacker breaks the encryption technique using all possible combinations of the secret key used for encryption. In the proposed LRMS, the LR encryption process requires a 256-bit determinant key and a 128-bit symmetric key. An attacker has to make 2^{384} guesses to hack the keys, which is computationally impossible. In addition, the attacker needs to guess the dynamic C2I table and operation code to gain access of plain LR. Hence, the proposed scheme is found to be resilient to brute-force attacks.

Corollary 3: The system thwarts a known-plaintext attack with a randomized and non-deterministic encryption process.

Proof:

In this attack, the attacker accesses the cipher text and its corresponding plain text. The target is to predict the encryption keys and develop a technique that can be used for decryption. Here, in the proposed LRMS, swap and XOR operations are selected based on a random determinant key, while the XOR operation is performed between the plain LR chunk and the symmetric key. Further, the character-to-integer mapping scheme is randomly determined, which makes the relation between cipher text and plain text non-deterministic. Besides, the attacker needs to guess the operation code to which is also randomly generated. Consequently, when encrypting two similar plain LRs, the system generates two distinct cipher LRs, enhancing its resilience against this specific type of attack.

Corollary 4: The LRMS prevents ciphertext-only attacks with dynamic and randomized elements.

Proof:

In this attack, the attacker cracks several encrypted messages. The goal of this attack is to retrieve as many plaintexts as feasible or to successfully infer the secret encryption keys. Once the encryption key is compromised, it becomes possible to decrypt all messages that have been encrypted using that specific key. In the proposed LRMS, the determinant key, symmetric key, C2I table, and operation code are generated randomly for every LR transmission. As a result, when decrypting two similar cipher LRs, the corresponding plain LRs are entirely distinct from each other. Thus, the proposed scheme can resist this attack.

Corollary 5: The system resists differential cryptanalysis attack with the determinant key, symmetric key, C2I mapping scheme, and operation code for each user interaction.

Proof:

In this attack, the attacker analyzes pairs of plaintext and their corresponding ciphertext with the goal of determining the encryption key, or more precisely, to reduce the time required to obtain the key. The proposed LRMS is secured against this attack as the determinant key, symmetric key, integer to character mapping scheme, and operation code are generated each time randomly for the same user. The same plain LR gets converted into a varying cipher LR during each LR transmission, and thus, the proposed scheme is resilient to this attack.

Corollary 6: The LRMS thwarts a man-in-the-middle attack by encrypting transactions.

Proof:

In this attack, the attacker seizes communications between the sender and receiver. This is either eavesdropping or modifying the traffic between two entities. The transactions and interactions among buyer, seller, bank, LRD, etc., are all encrypted, while the determinant-random key, symmetric-random key, C2I table, and operation are encrypted using the public key before they are sent to the entity. Therefore, the authorized entity can only retrieve the information by using its own private key. As the private key is only known to the authorized entity, no one except the can retrieve the details. Note that the proposed scheme adopts the ElGamal asymmetric cryptosystem, which is a highly secure encryption technique. Thus, the proposed scheme survives this attack.

Corollary 7: The system combats phishing attacks with secure registration, encryption, and precise data exchange processes.

Proof:

In this attack, a malicious user retrieves the personal information of the receiver. Unauthorized access to the receiver’s sensitive information can potentially enable an attacker to gain access to the sender’s data. In the proposed LRMS, the patient and authorized entity have to register with the system manager. After completing the registration, the system manager sends the public and private key pair to the patient and authorized entity. Then, during e-MR access, the authorized entity needs to send a request to the patient in an encrypted form. The authorized entity gets cipher e-MR from the system manager after a successful authentication. Then the patient sends the secret key to the authorized entity. The patient does not ask the authorized entity for any personal information. So, it is not possible for the malicious user to get access to the required information from the authorized entity, and thus, the proposed scheme resists the phishing attack.

Corollary 8: The LRMS demonstrates a robust avalanche effect subject to a 1-bit flipping of the determinant key.

Proof:

The avalanche effect is a very important security measure of an encryption technique, which means that a small change in the input (either plaintext or key) results in a significant change in the output (cipher text). The avalanche effect ensures that the output of a system is completely unpredictable and that even a small change in the input produces a completely different output. This makes it very difficult for an attacker to determine the input from the output, which is necessary for breaking the encryption. Any encryption technique that fails to exhibit enough avalanche effect may be vulnerable to attacks such as brute force, differential cryptanalysis, etc. An encryption technique with a 50% or more avalanche effect is considered good. The avalanche effect can be calculated using Eq. (3), and the results are given in Table 7. It shows that the minimum value is 55.5% and the maximum value is 60.2% when 1-bit flipping of the 256-bit determinant key.

$$\text{Avalanche effect} = \frac{\text{\# of bits change in ciphertext}}{\text{\# of bits in ciphertext}} \times 100 \quad (3)$$

Corollary 9: The cryptosystem passed all NIST tests, confirming its robustness and suitability for cryptography.

Proof:

It is shown that the hybrid chaotic map passes all the statistical tests defined by NIST presented in Table 3.

Table 7 Analysis of the avalanche effect.

Input size (KB)	Change in cipher land record
1	60.2%
5	55.5%
10	58.8%

As the proposed encryption technique works based on the hybrid chaotic map, it is obvious that it will also pass all the statistical tests. Table 8 presents the statistical analysis report of the proposed encryption technique.

Table 8 Nist randomness test analysis for encryption technique.

Sl. No.	Test name	*P* value	Result
1	Frequency (Monobit)	0.676	Pass
2	Frequency (Block)	0.271	Pass
3	Run	0.188	Pass

Sl. No.	Test name		P value	Result
4	Longest run		0.277	Pass
5	Binary matrix rank		0.693	Pass
6	Discrete Fourier transform (spectral)		0.852	Pass
7	Non-overlapping template matching		0.611	Pass
8	Overlapping template matching		0.021	Pass
9	Universal		0.334	Pass
10	Linear complexity		0.020	Pass
11	Serial cumulative sums	Value 1	0.723	Pass
		Value 2	0.554	Pass
12	Approximate entropy		0.445	Pass
13	Cumulative sums	Forward	0.807	Pass
		Reverse	0.528	Pass
14	Random excursions (State: -4)		0.859	Pass
15	Random excursions variant (State: -4)		0.735	Pass

7. Complexity analysis

This section analyzes the time complexity of encryption, decryption, key generation, and key retrieval algorithms for the proposed LRMS and compares them with four other existing systems shown in Table 9. It confirms that the encryption and decryption time complexities of the proposed system outperform the work proposed in [31]. Furthermore, the developed LRMS exhibits lower time key generation complexity compared to the study proposed in Refs. [32] and [33]. On the other hand, the system in Ref. [34] employs an asymmetric cryptosystem in the encryption process, which requires very high time complexity. The overall time complexity of the proposed scheme is O(n), which is similar to the scheme [35]. Nevertheless, the proposed LRMS provides higher security compared to [35] because of the involvement of a dynamic C2I table, hybrid chaotic map-based key generation, requirement of two keys for encryption and three keys for decryption.

Table 9 Time complexity analysis.

		Scheme					
		[31]	[32]	[33]	[34]	[35]	Our
Phase	Key generation	O(n)	O(n^2)	O(n^2)	O(n^2)	O(n)	O(n)
	Encryption	O(n^2)	O(n)	O(n)	O(n^2)	O(n)	O(n)
	Decryption	O(n^2)	O(n)	O(n)	O(n^2)	O(n)	O(n)

8. Experimental studies

8.1. Experimental setup

The prototype of the proposed LRMS is developed on Windows 11 OS utilizing Intel(R) Core™ i5-7300HQ CPU @2.50 GHz 64-bit processor with 8GB of RAM. It was developed in Visual Studio Code 2019. The ElGamal cryptosystem was employed as an asymmetric cryptosystem, utilizing a 1024-bit key length for operations. In order to conduct real-life testing, this paper utilized the web3.storage (IPFS) [36] as a distributed storage server and Sepolia Ethereum test network [37] as blockchain storage.

8.2. Output of the encryption phase

Considering the plain LR 'Seller: Alice, Buyer: Bob, Land information: Plot number: 200/A, Land certificate: 50, Area:150D, Transaction ID: TxFYXY23F45' and using the steps of the Section IV(D-1), Table 10 depicts the output of the encryption phase.

8.*3. Output of the decryption phase*

Using the steps of the section IV(D-2) and by considering the output of section VIII(B) as input, Table 11 presents the output of the decryption phase.

Table 10 Output of the Encryption Phase.

<table>
<tr><td>Step</td><td>Operation</td><td colspan="3">Output</td></tr>
<tr><td>Step 1</td><td>Read land data</td><td colspan="3">Seller: Alice, Buyer: Bob, Land information: Plot number: 200/A, Land certificate: 50, Area:150D, Transaction ID: TxFYXY23F45</td></tr>
<tr><td>Step 2</td><td>Generate a dynamic C2I table</td><td colspan="3">A:32, x:37, T:72, 6:07, c:14,…</td></tr>
<tr><td>Step 3</td><td>Convert the text data into the corresponding integer value</td><td colspan="3">25476637463341547863547646312579414848633574632786324063192375032361415450425[*]</td></tr>
<tr><td rowspan="2">Step 4</td><td>Make equal-length chunk</td><td colspan="3">233415475472106637468635476[*], 1586354764631257941484[*], 5848486335746327863240[*],…</td></tr>
<tr><td>Convert into bits</td><td colspan="3">0101101010101011101100[*], 011010101001111101010100[*], 0100110101001111101010100[*],…</td></tr>
<tr><td>Step 5</td><td>Set variable m and r to 1</td><td colspan="3">m = 1 and r = 1</td></tr>
<tr><td>Ste</td><td>Repeat Step</td><td>Pass 1</td><td>Pass 2</td><td>Pass k</td></tr>
</table>

Step	Operation	Output		
p 6	6 to Step 10 until unencrypted chunks available			
Step 7	Scan *m*th and (256 – *m*)th bit of the determinant key	1st bit = 0 and 255th bit = 0 (Code: 00)	2nd bit = 1 and 254th bit = 0 (Code: 10)	…
Step 8	Select operations based on two bits	Op1: *chunk* #1 ⊕ key_d	Op1: *chunk* #1 ⊕ key_s	…
		Op2: *chunk* #2 ⊕ key_d	Op2: *chunk* #2 ⊕ key_d	
		Op3: No	Op3: Yes	
Step 9	Perform the selected operations to *r*th and/or (*r*+1)th chunk using the	1st chunk: 0110101100* (*chunk* #1⊕key_d)	3rd chunk: 01100101111* (*chunk* #1⊕key_s)	…
		2nd chunk: 010001011* (*chunk* #1⊕key_d)	4th chunk: 01010101010* (*chunk* #1⊕key_d)	…

Step	Operation	Output				
	key_s or key_d					
	Swap positions of the chunks (if necessary)	1st chunk: 0110101100*	No swap	3rd chunk: 01010101010*	Swap	…
		2nd chunk: 010001011*		4th chunk: 01100101111*		…
Step 10	Increase the value of m and r	$m = 2$, $r = 2$	$m = 3$, $r = 3$			$m = x$, $r = y$
Step 11	Merge the chunks to form a single encrypted chunk	1000011001100100011101011101011111101101100101111111010100000101110011100101001110110101110101100010001001100101111100110111001010000100111110*				
Step 12	Convert each 6-bits into its corresponding integer value	54786354764631225454649985654645776889800923750663746334157932361415450425633574*				
Step	Replace each	si a%rchitect#o AFJTJ% dFcta s88nt jNemo enim volu!ptas@ sit ipsam# volu5ptatem66 quia*				

Step	Operation	Output
13	integer with its corresponding character	

[*] denotes a portion of data

Table 11 Output of the retrieval phase.

Step	Operation	Output
Step 1	Read cipher data	si a%rchitect#o AFJTJ% dFcta s88nt jNemo enim volu!ptas@ sit ipsam# volu5ptatem66 quia[*]
Step 2	Get m, r and dynamic C2I table, determinant key and symmetric key	$m = x$, $r = y$
		A:32, x:37, T:72, 6:07, c:14,…
		Determinant key: 0110101110111010111010101001100101111100110111001010000100111110[*]
		Symmetric key: 1101011101011101011000100010011001011111001101110010100001001111 10[*]

<table>
<tr><th>Step</th><th>Operation</th><th colspan="3">Output</th></tr>
<tr><td>Step 3</td><td>Replace each character to its corresponding integer value</td><td colspan="3">5478635476463122545464998565464577688980092375066374633415793236141545042563357 4*</td></tr>
<tr><td>Step 4</td><td>Convert each integer into its corresponding 6-bit binary</td><td colspan="3">01101011000101010111011001010111111011011001011111101010000010111001110010100111011010111010110001000100110010111110011011100101000010011111 0*</td></tr>
<tr><td>Step 5</td><td>Make equal-length chunk</td><td colspan="3">0110101100010101011101100*, 0100010111001111101010100*, 0101010101001111101010100*,…</td></tr>
<tr><td>Step 6</td><td>Repeat Step 6 to Step 10 until unencrypted chunks available</td><td>Pass 1</td><td>Pass 2</td><td>Pass k</td></tr>
</table>

Step	Operation	Output		
Step 7	Scan *m*th and (256 – *m*)th bit of the determinant key	1st bit = 0 and 255th bit = 0 (Code: 00)	2nd bit = 1 and 254th bit = 0 (Code: 10)	…
Step 8	Select the operations based on two bits	Op1: *chunk* #1 ⊕ key_d	Op1: *chunk* #1 ⊕ key_s	…
		Op2: *chunk* #2 ⊕ key_d	Op2: *chunk* #2 ⊕ key_d	
		Op3: No	Op3: Yes	
Step 9	Perform the selected operations to *r*th and/or (*r* + 1)th chunk using the key_s or key_d	1st chunk: 010110101* (*chunk* #1⊕key_d)	3rd chunk: 0111010101* (*chunk* #1⊕key_s)	…
		2nd chunk: 011010101* (*chunk* #1⊕key_d)	4th chunk: 0100110101* (*chunk* #1⊕key_d)	…
	Swap positions of the	1st chunk: 010110101* / No swap	3rd chunk: 01001101010* / Swap	…

Step	Operation	Output				
	chunks (if necessary)	2nd chunk: 011010101*		4th chunk: 0111010101*		…
Step 10	Decrease the value of *m* and *r*	$m = x – 1$, $r = y – 1$		$m = x – 2$, $r = y – 2$		$m = 1$, $r = 1$
Step 11	Merge all chunks to form a single binary chunk	010110101001100110010001110101110101111110110110010111111101010000010111001110010100111011010111010110001000100110010111110011010101000000111010 01111*				
Step 12	Convert into integer chunk	23341547547210663746863547615863547646312579414845848486335746327863240511 52151*				
Step 13	Convert each 2-ineger-digit into corresponding character and get	Seller: Alice, Buyer: Bob, Land information: Plot number: 200/A, Land certificate: 50, Area:150D, Transaction ID: TxFYXY23F45				

Step	Operation	Output
	the plain land record	

[*] denotes a portion of data

8.4. Efficiency analysis

The efficiency of the proposed system is shown in this section. Table 12 presents the time required for key generation using the HCS with varying key sizes. Each test has been repeated 50 times to ensure reliability. In addition, Table 13 provides insights into the time requirements for both encryption and decryption operations of the proposed lightweight cryptosystem. Both tables collectively demonstrate the remarkable efficiency of the proposed LRMS, as they demand notably minimal time requirements for these operations. Besides, the plaintext and ciphertext ratio is another concern for the effectiveness of a cryptosystem. The proposed cryptosystem generates almost 1:1 (plain LR: cipher LR) as presented in Table 14, which makes the proposed LRMS efficient. Moreover, the cipher LR uploading time requirement in IPFS is cheap, as shown in Fig. 13. Furthermore, Fig. 14 shows a transaction snippet held in Sepolia Ethereum test network.

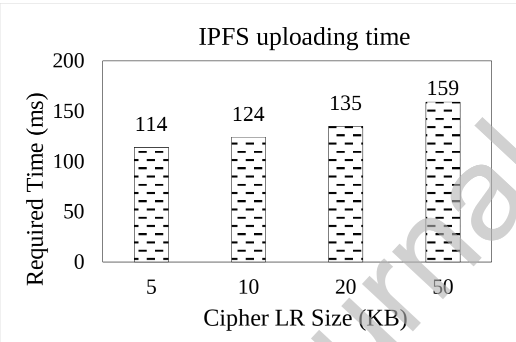


Fig. 13. Land record uploading time for different size in IPFS.

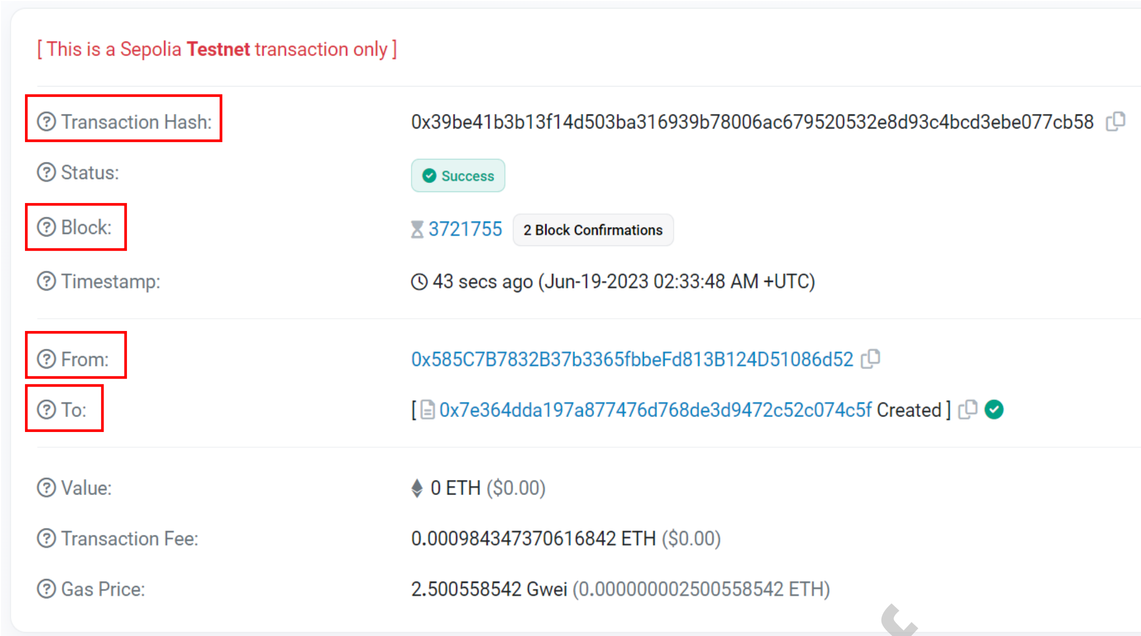


Fig.14. Sepolia test network transaction snippet.

8.5. Comparison with other LRMS

The proposed system exhibits significant advantages over the state-of-the-art systems in key aspects such as data integrity, privacy, trading capabilities, lightweight cryptography, encryption key distribution, multi-level security, and scalability, as summarized in Table 15. Unlike most other systems that offer only basic security protocols, our LRMS implements multi-level security, combining layered cryptographic measures to provide a robust LR privacy shielding against unauthorized access. The table also shows that the proposed LRMS offers lower text-to-integer conversion overhead along with dynamic conversion, which can contribute to more efficient and adaptable operations. Moreover, the dynamic conversion capability and multi-level security measures not only bolster operational efficiency but also provide a higher level of protection for sensitive LRs. In contrast to most other systems, which rely on a trusted land registration department (LRD), our LRMS operates with a semi-trusted LRD, offering a more flexible trust model that reduces dependency on a single trusted entity. Additionally, the incorporation of dynamic encryption key distribution bolsters the robustness of the LRMS. Our LRMS also features a decentralized advertising platform for land transactions, a component missing in most other works, which simplifies land trading by allowing landowners to post listings and prospective buyers to search securely. Furthermore, the exploitation of IPFS for LR storage and lower text-to-integer conversion overhead ensures the scalability of the proposed LRMS. The proposed system's specifications align with modern hardware and software requirements, ensuring compatibility and operational efficiency. So, Table 15 highlights the proposed LRMS's superior suitability for contemporary LR management needs, presenting it as a robust, scalable, and secure alternative to state-of-the-art systems.

Table 12 Key generation time.

Keys Size (bits)	Time (ms)
128	0.0036

Keys Size (bits)	Time (ms)
256	0.0039
512	0.0043
1024	0.0056

Table 13 Encryption & decryption time.

Data Size (KB)	Time (ms)	
	Encryption	Decryption
5	0.0104	0.0091
10	0.0205	0.0191
20	0.0409	0.0412
50	0.1032	0.0986

Table 14 Plain LR size vs. cipher LR size.

Plain LR size (KB)	Cipher LR size (KB)
5	5.24
10	10.29
20	20.33
50	50.42

Table 15 Comparison with other systems.

Work	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13
[5]	√	×	×	Trusted	×	×	Normal	Static	×	√	–	–	Ethereum & Hybrid
[6]	√	×	×	Trusted	×	×	Normal	Static	×	×	–	–	Ethereum
[7]	√	×	×	Trusted	×	×	Normal	Static	×	×	–	–	Ethereum
[10]	√	×	×	Trusted	×	×	Normal	Static	×	×	–	–	–
[11]	√	√	×	Trusted	×	×	Normal	Static	×	×	–	–	–
[12]	√	×	×	Trusted	×	×	Normal	Static	×	√	–	–	–
[13]	√	×	×	Trusted	×	×	Normal	Static	×	×	–	–	–
[14]	√	×	×	Trusted	×	×	Normal	Static	×	√	–	–	–
[15]	√	×	×	Trusted	×	×	Normal	Static	×	√	Core i7–7700, 3.6 GHz, 8GB RAM	Windows OS, IPFS	–
[16]	√	×	×	Trusted	×	×	Normal	Static	×	√	Xeon X5355, 2.66 GHz,	Windows OS, IPFS	–

Work	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13
											8GB RAM		
[2]	√	√	Centralized	Trusted	×	×	Less	Static	×	√	Core i5-7300, 2.50 GHz, 8GB RAM	Windows OS, IPFS	Ethereum
[17]	√	√	×	Trusted	×	×	Normal	Static	×	√	Core i7-1165 G7, 2.80 GHz, 16.0 GB RAM	Windows OS, IPFS	Ethereum
[18]	√	√	×	Trusted	×	×	Normal	Static	×	√	–	–	–
[19]	√	×	×	Trusted	×	×	Normal	Static	×	√	–	–	–
[20]	√	×	×	Trusted	×	×	Normal	Static	×	×	–	–	Hyperledger
[21]	√	×	×	Trusted	×	×	Normal	Static	×	×	–	Linux OS, Docker	Hyperledger
[22]	√	√	×	Trusted	×	×	Normal	Static	×	×	–	Linux OS, EVE-NG	Hyperledger

Work	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13
[23]	√	×	×	Trusted	×	×	Normal	Static	×	×	–	–	Ethereum
Proposed	√	√	Decentralized	Semi Trusted	√	√	Less	Dynamic	√	√	Core i5-7300, 2.50 GHz, 8GB RAM	Windows OS, IPFS	Ethereum

9. Discussion and conclusion

This paper introduces a blockchain-based LRMS aimed at enhancing the lightweight & multi-level security and efficiency of LR management. The system prioritizes critical aspects such as LR privacy, integrity, etc. of along with the scalability of LRMS. Leveraging a lightweight cryptosystem, it fortifies LR privacy, ensuring that sensitive information remains confidential. Multi-level security is achieved through the incorporation of the semi-trusted LRD, the lightweight cryptosystem, the encryption key distribution, and the dynamic character-to-integer mapping. Additionally, by storing LRs on the decentralized platform, the system guarantees the integrity of the LR, minimizing the risk of tampering or unauthorized alterations. Additionally, the LRMS also offers a decentralized trading platform, simplifying the process of locating and transacting land parcels, thereby reducing the administrative burden associated with land searches. Besides, its introduction of the dynamic C2I table, which optimizes text to integer conversion by reducing conversion overhead by an impressive 33% when compared to traditional ASCII tables, along with a level of confusion.

As LRMS is append only solution, the system becomes more cumbersome, potentially slowing down. Several prominent blockchain scaling techniques (e.g., on-chain solutions like sharding, increased block size, improved consensus protocol, etc., and off-chain solutions like state channel, sidechain, plasma, validium, rollups, etc.) have been proposed to address issues like blockchain bloat, transaction throughput, high transaction fee, network efficiency, etc. While on-chain solutions have a direct effect on the main blockchain architecture, the off-chain solutions do not. In the case of LRMS, the sharding technique could be particularly beneficial for large-scale land registries, where geographical regions or administrative divisions (e.g., districts, counties, states) could be mapped to different shards. Each shard could handle a subset of land transactions, improving scalability by parallelizing transaction processing. However, it is important to address the complexity of cross-shard communication to maintain consistency, especially for transactions that span multiple regions. Besides, the sidechain technique could be used to store metadata or historical land data that does not require frequent updates or querying. That means the detailed LR could be stored in the sidechain, while the LR metadata remains on the main blockchain. However, we cannot simultaneously achieve decentralization, security, and scalability at a time, which is called the scaling trilemma.

The proposed LRMS employed the IPFS to ensure decentralized data access but lacks guaranteed persistence, as files may be lost if not actively pinned or retained by nodes. Network reliability is also variable, affecting data availability during node downtime or high latency periods. Future work will address these issues by integrating Filecoin to incentivize data retention, ensure long-term storage.

Alternatives like Arweave, with its focus on permanent storage, and Storj, offering high redundancy and global distribution, will be evaluated for cost, scalability, and performance.

The assessment of this proposed scheme demonstrates its effectiveness in addressing the identified challenges in the LRMS. Future plans include further enhancements to the encryption and decryption processes, as well as the implementation of a working prototype of the LRMS on a large scale and more realistic operational environment. This forward-thinking approach indicates a commitment to practical implementation and continual improvement, promising a more secure and efficient LRMS for the future.

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CRediT author statement

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: